

Problem 4.16

Consider an investment with nonzero interest rate i . If i_5 is equal to i_{10} , show that interest is not computed using simple interest.

Solution

If the investment were using simple interest, then the accumulation function would be

$$a(t) = 1 + it.$$

The effective rate of interest in period n is

$$i_n = \frac{a(n) - a(n-1)}{a(n-1)}.$$

So,

$$i_5 = \frac{a(5) - a(4)}{a(4)} \quad \text{and} \quad i_{10} = \frac{a(10) - a(9)}{a(9)}.$$

Under simple interest,

$$a(5) = 1 + 5i, \quad a(4) = 1 + 4i, \quad a(10) = 1 + 10i, \quad a(9) = 1 + 9i.$$

Hence,

$$i_5 = \frac{(1 + 5i) - (1 + 4i)}{1 + 4i} = \frac{i}{1 + 4i},$$

and

$$i_{10} = \frac{(1 + 10i) - (1 + 9i)}{1 + 9i} = \frac{i}{1 + 9i}.$$

If $i_5 = i_{10}$, then

$$\frac{i}{1 + 4i} = \frac{i}{1 + 9i}.$$

Since the interest rate is nonzero, $i \neq 0$, so we may cancel i and get

$$\frac{1}{1 + 4i} = \frac{1}{1 + 9i}.$$

Therefore,

$$1 + 4i = 1 + 9i,$$

which implies

$$4i = 9i.$$

So,

$$5i = 0,$$

and hence

$$i = 0.$$

But this contradicts the assumption that the interest rate is nonzero.

Therefore, interest cannot be computed using simple interest.